

Chapter 4: Model Checking in the Cox Regression Model

Modelling Survival Data in Medical Research

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Part 1

1. Why Model Checking Matters
2. Types of Residuals
 - ▷ Cox-Snell residuals
 - ▷ Modified Cox-Snell residuals
 - ▷ Martingale residuals
 - ▷ Deviance residuals
 - ▷ Schoenfeld residuals
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3. Example 4.1: Kidney Infection Data
4. Assessment of Model Fit
 - ▷ Cox-Snell cumulative hazard plot
 - ▷ Martingale & deviance residual plots
 - ▷ Checking functional form of covariates
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Why Is Model Checking Essential?

After fitting a Cox regression model, we must ask: **is the model actually appropriate?**

What can go wrong:

- ▶ Wrong set of explanatory variables included
- ▶ Wrong functional form (e.g. linear when non-linear)
- ▶ One or two observations dominate the fit
- ▶ Proportional hazards (PH) assumption is violated

Why visual inspection alone is not enough:

- ▶ Many variables $\Rightarrow$ high-dimensional — hard to spot by eye
- ▶ Censoring obscures patterns in simple plots

The model-checking workflow

- 1 Compute **residuals** for each individual
- 2 Plot residuals to detect inadequacies
- 3 Identify **influential observations**
- 4 Test the **proportional hazards assumption**
- 5 Revise the model if needed

Model checking is not optional — it is an essential part of the modelling process.

What We Are Checking

Functional form

Is each covariate included on the right scale?

Should we use X or $\log X$ or X^2 ?

Tool: martingale residuals vs. covariate

Influential observations

Does any single patient drive the results?

Would conclusions change if we removed them?

Tools: delta-betas, $|\ell_{\max}|$, likelihood displacement

Proportional hazards

Is the hazard ratio constant over time, or does it change?

Tools: log-cumulative hazard plot, Schoenfeld residuals, Grambsch-Therneau test

Residuals are the common thread: they are quantities computed for each individual whose **behaviour is known** when the fitted model is correct.

Setup: The Fitted Cox Model

Throughout Chapter 4, assume a Cox model has been fitted to n individuals with r deaths:

$$\hat{h}_i(t) = \exp(\hat{\beta}' \mathbf{x}_i) \hat{h}_0(t)$$

For each individual i :

- ▶ t_i : observed survival time (possibly censored)
- ▶ δ_i : event indicator (1 = death, 0 = censored)
- ▶ $\hat{\eta}_i = \hat{\beta}' \mathbf{x}_i$: risk score
- ▶ $\hat{H}_0(t_i)$: estimated baseline cumulative hazard at t_i
- ▶ $\hat{H}_i(t_i) = e^{\hat{\eta}_i} \hat{H}_0(t_i)$: individual cumulative hazard
- ▶ $\hat{S}_i(t_i) = e^{-\hat{H}_i(t_i)}$: individual survivor function

Six types of residuals defined in this chapter:

Residual	Main use
Cox-Snell r_{Ci}	Overall fit
Modified CS r'_{Ci}	Adjust for censoring
Martingale r_{Mi}	Functional form
Deviance r_{Di}	Outlier detection
Schoenfeld r_{Sji}	PH assessment
Score r_{Uji}	Influence diagnostics

Cox-Snell Residuals: Definition

Cox-Snell residual for individual i

$$r_{Ci} = \exp(\hat{\beta}' \mathbf{x}_i) \hat{H}_0(t_i) = \hat{H}_i(t_i) = -\log \hat{S}_i(t_i)$$

The estimated cumulative hazard experienced by individual i up to their observed time t_i .

Key theoretical property:

If T has survivor function $S(t)$, then $Y = -\log S(T)$ has an **exponential distribution with unit mean**.

Therefore: if the fitted model is correct, the Cox-Snell residuals r_{Ci} behave like a **censored sample from a unit exponential distribution**.

Properties:

- ▶ Always positive (cannot be negative)
- ▶ Not symmetric about zero
- ▶ Highly skewed (exponential-like)
- ▶ Mean and variance both ≈ 1 under correct model
- ▶ Censored observations $\Rightarrow$ corresponding residuals are also censored

Why Residuals Should Follow a Unit Exponential

The key mathematical result (proof sketch):

Let T be a survival time with survivor function $S(t)$, and define $Y = -\log S(T)$. Then:

Using the change-of-variables formula for densities:

$$f_Y(y) = f_T(S^{-1}(e^{-y})) \left/ \left| \frac{dy}{dt} \right| \right|$$

Since $\frac{dy}{dt} = \frac{f_T(t)}{S(t)}$, substituting gives:

$$f_Y(y) = e^{-y}, \quad y > 0$$

which is the density of a **unit exponential**.

Why this matters for model checking:

If the correct model is fitted:

$$\hat{S}_i(t_i) \approx S_i(t_i)$$

So $-\log \hat{S}_i(t_i) = r_{Ci}$ should behave like unit exponential observations.

Deviations from a unit exponential suggest model inadequacy.

Modified Cox-Snell Residuals

Problem: censored residuals are systematically underestimates, the patient survived longer than their censoring time, so their true residual would be larger.

Solution: add an excess residual Δ to censored observations.

Using the **lack-of-memory property** of the exponential distribution: given survival to time t_0 , the *additional* survival time is also exponential with the same mean. So Δ has a unit exponential distribution, with mean = 1 and median = $\log 2 = 0.693$.

Version 1 (mean correction):

$$r'_{Ci} = \begin{cases} r_{Ci} & \text{if uncensored } (\delta_i = 1) \\ r_{Ci} + 1 & \text{if censored } (\delta_i = 0) \end{cases}$$

Or: $r'_{Ci} = 1 - \delta_i + r_{Ci}$

Version 2 (median correction, Crowley-Hu):

$$r''_{Ci} = \begin{cases} r_{Ci} & \text{if uncensored} \\ r_{Ci} + 0.693 & \text{if censored} \end{cases}$$

Crowley and Hu (1977) found that adding 1 inflates the residual too much; 0.693 is more conservative.

Martingale Residuals

Definition

$$r_{Mi} = \delta_i - r_{Ci} = \delta_i - \hat{H}_i(t_i)$$

Observed deaths minus expected deaths for individual i over $(0, t_i)$.

Properties:

- ▶ Range: $(-\infty, 1]$
- ▶ Sum to zero: $\sum_i r_{Mi} = 0$
- ▶ Uncorrelated in large samples, $E(r_{Mi}) = 0$
- ▶ Censored observations have $\delta_i = 0$, so $r_{Mi} = -r_{Ci} < 0$

Interpretation:

- ▶ $r_{Mi} \approx +1$: individual died *sooner* than the model predicted — unexpectedly short survival
- ▶ $r_{Mi} \ll 0$: individual *lived longer* than the model predicted — unexpectedly long survival
- ▶ Residuals near zero: well-fitted observations

Limitation: martingale residuals are *skewed* (range $(-\infty, 1]$ is not symmetric), making plots harder to interpret.

Definition

$$r_{Di} = \text{sgn}(r_{Mi}) \left[-2 \{ r_{Mi} + \delta_i \log(\delta_i - r_{Mi}) \} \right]^{1/2}$$

where $\text{sgn}(x) = +1$ if $x > 0$ and -1 if $x < 0$.

Why deviance residuals?

Martingale residuals are skewed on $(-\infty, 1]$. The deviance transformation:

- ▶ Shrinks large negative martingale residuals (dominant r_{Mi} term)
- ▶ Expands residuals near +1 (dominated by $\log(1 - r_{Mi}) \rightarrow -\infty$)
- ▶ Result: approximately **symmetric about zero**

Properties:

- ▶ More symmetric than martingale residuals
- ▶ Same sign as the corresponding martingale residual
- ▶ Do **not** necessarily sum to zero
- ▶ $D = \sum_i r_{Di}^2$ is the model deviance
- ▶ Large $|r_{Di}| \Rightarrow$ individual not well fitted

Prefer deviance residuals for index plots and outlier detection.

Definition (for variable X_j , individual i)

$$r_{Sji} = \delta_i(x_{ji} - \hat{a}_{ji}), \quad \hat{a}_{ji} = \frac{\sum_{l \in R(t_i)} x_{jl} e^{\hat{\beta}' \mathbf{x}_l}}{\sum_{l \in R(t_i)} e^{\hat{\beta}' \mathbf{x}_l}}$$

$\hat{a}_{ji}$ is the **weighted mean of X_j** in the risk set at time t_i .

Key features:

- ▶ One residual *per variable per death time* (not per individual)
- ▶ Only non-zero for uncensored observations ($\delta_i = 1$)
- ▶ Zero for the largest uncensored time
- ▶ Sum to zero for each variable
- ▶ Do not depend on $\hat{H}_0(t)$

Scaled Schoenfeld residuals

(Grambsch-Therneau):

$$\mathbf{r}_{Si}^* = d \widehat{\text{var}}(\hat{\beta}) \mathbf{r}_{Si}$$

d = total deaths, $\widehat{\text{var}}(\hat{\beta})$ = vcov matrix.

Used for testing proportional hazards (Section 4.4).

Definition (for variable X_j , individual i)

$$r_{Uji} = r_{Sji} + e^{\hat{\beta}' \mathbf{x}_i} \sum_{t_r \leq t_i} \frac{(\hat{a}_{jr} - x_{ji}) \delta_r}{\sum_{l \in R(t_r)} e^{\hat{\beta}' \mathbf{x}_l}}$$

The score residual is a **modification of the Schoenfeld residual**.

Properties:

- ▶ One value per variable per individual
- ▶ Non-zero even for censored observations
- ▶ Sum to zero for each variable
- ▶ Contribution of individual i to the efficient score for β_j , using only information up to time t_i

Main uses:

Score residuals are the building blocks for **influence diagnostics**:

- ▶ Delta-beta approximations (Section 4.3.1)
- ▶ Likelihood displacement (Section 4.3.2)
- ▶ $|\ell_{\max}|$ diagnostic (Section 4.3.2)

Summary: Six Residuals at a Glance

Residual	Symbol	Range	Sum to 0?	Primary use
Cox-Snell	r_{Ci}	$(0, \infty)$	No	Overall fit (unit exp. plot)
Modified Cox-Snell	r'_{Ci}	$(0, \infty)$	No	Adjust censored CS residuals
Martingale	r_{Mi}	$(-\infty, 1]$	Yes	Functional form, outliers
Deviance	r_{Di}	$(-\infty, \infty)$	No	Outliers, index plots
Schoenfeld	r_{Sji}	$(-\infty, \infty)$	Yes	PH assessment
Score	r_{Uji}	$(-\infty, \infty)$	Yes	Influence diagnostics

All residuals described above are computed automatically in R via `residuals(cox.fit, type = "...")` with types "martingale", "deviance", "schoenfeld", "score", "dfbeta", "scaled.schoenfeld".

Example 4.1 — Kidney Infection Data

Study: Infection in patients on dialysis (McGilchrist & Aisbett, 1991)

13 kidney patients. Time from catheter insertion to removal (due to infection or other reason).

Patient	Time	Status	Age / Sex
1	8	1	28 / M
2	15	1	44 / F
3	22	1	32 / M
4	24	1	16 / F
5	30	1	10 / M
6	54	0	42 / F
7	119	1	22 / F
8	141	1	34 / F
9	185	1	60 / F
10	292	1	43 / F
11	402	1	30 / F
12	447	1	31 / F
13	536	1	17 / F

Fitted Cox model:

$$\hat{h}_i(t) = e^{0.030 \text{ Age}_i - 2.711 \text{ Sex}_i} \hat{h}_0(t)$$

(Sex: 1 = male, 2 = female)

Model notes:

- ▶ Sex is significant ($P = 0.011$): females have much lower hazard of infection
- ▶ Age is not significant ($P = 0.251$) but is retained for clinical reasons
- ▶ Only 1 censored observation (patient 6)

Example 4.1 — Residuals Table (Selected)

Pt.	r_{Ci}	r_{Mi}	r_{Di}	r_{S1i}	r_{S2i}	r_{S1i}^*	r_{S2i}^*	r_{U1i}	r_{U2i}
1	0.280	0.720	1.052	-1.085	-0.242	0.033	-3.295	-0.781	-0.174
2	0.072	0.928	1.843	14.493	0.664	0.005	7.069	13.432	0.614
3	1.214	-0.214	-0.200	3.129	-0.306	0.079	-4.958	-0.322	0.058
4	0.084	0.916	1.765	-10.222	0.434	-0.159	8.023	-9.214	0.384
5	1.506	-0.506	-0.439	-16.588	-0.550	-0.042	-5.064	9.833	0.130
6*	0.265	-0.265	-0.728	—	—	—	—	-3.826	-0.145
7	0.235	0.765	1.168	-17.829	0.000	-0.147	3.083	-15.401	-0.079
...	...	...	...	...	...	...	...	...	...
13	2.195	-1.195	-0.904	0.000	0.000	0.000	0.000	16.246	-0.068

*Patient 6 is censored. Schoenfeld residuals undefined (shown as —).

Patient 13 has the longest time: $r_{Sji} = 0$ (by definition).

Note the skewness of r_{Ci} (range 0.07–2.20) vs. near-symmetry of r_{Di} (range -0.9 to +1.8).

Martingale, Schoenfeld, and score residuals all sum to zero.

Cox-Snell Residual Plot: The Idea

If the model is correct: $r_{Ci} \sim \text{Exp}(1)$ (unit exponential, censored).

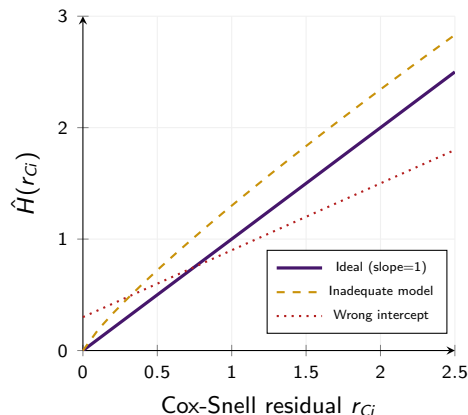
For a unit exponential, the cumulative hazard is:

$$H(t) = -\log S(t) = -\log(e^{-t}) = t$$

So a plot of $H(t)$ vs. t is a **straight line through the origin with slope 1**.

Procedure:

- 1 Compute residuals r_{Ci}
- 2 Obtain KM estimate $\hat{S}(r_{Ci})$ treating r_{Ci} as survival times
- 3 Plot $\hat{H}(r_{Ci}) = -\log \hat{S}(r_{Ci})$ against r_{Ci}
- 4 Should give a straight line with slope 1 and intercept 0



Cox-Snell Plot: Limitations

The Cox-Snell residual plot looks appealing in theory, but has serious practical limitations:

Why the plot can mislead

- ▶ The r_{Ci} are *estimated*, not true values. Estimating $\hat{\beta}$ and $\hat{H}_0(t)$ distorts the distribution away from unit exponential.
- ▶ For small n , the distribution can be very different from unit exponential (Lagakos, 1981 showed this for $n = 3$).
- ▶ Even when the model is *known to be wrong*, the plot often still looks like a straight line with slope 1.
- ▶ The null model (no covariates) always gives a straight line — even when covariates should be included.

Practical conclusion:

Use the Cox-Snell plot as a *quick sanity check*, not as a reliable model adequacy test.

A straight-line plot does **not** confirm the model is correct.

A non-straight-line plot does suggest serious problems.

Prefer plots based on martingale or deviance residuals for assessing model fit.

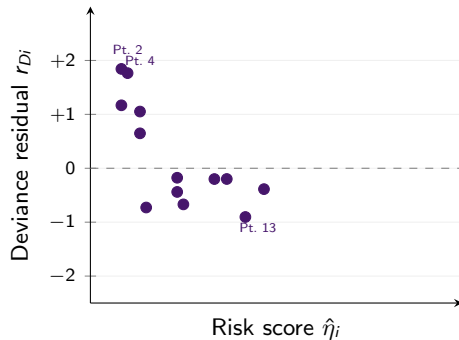
Martingale Residual Plots: Index Plot and vs. Risk Score

Index plot (residual vs. observation number):

- ▶ Highlights individuals whose survival time is *not well fitted*
- ▶ Large positive $r_{Mi} \approx 1$: died unexpectedly soon
- ▶ Large negative r_{Mi} : lived unexpectedly long
- ▶ These are **outliers**

Plot vs. risk score $\hat{\eta}_i$:

- ▶ The risk score summarises all covariate information
- ▶ Large $|r_{Di}|$ with low risk score: patient at low risk but died sooner than expected
- ▶ Combines outlier detection with covariate context



Deviance vs. Martingale Residuals: When to Use Which

Martingale residuals

- ▶ Range: $(-\infty, 1]$ — asymmetric
- ▶ Best for: plotting against *covariates* to assess functional form
- ▶ The shape of the scatter reveals the relationship needed between hazard and covariate
- ▶ Smoother superimposed helps identify whether a linear, log, or other term is needed

Deviance residuals

- ▶ Range: approximately symmetric around 0
- ▶ Best for: *index plots* and plots against risk score to find outliers
- ▶ Values $|r_{Di}| > 2$ suggest potential outliers
- ▶ Prefer these over martingale residuals for visual displays

For functional form checking: use **martingale residuals from the null model** plotted against each covariate.

For outlier detection: use **deviance residuals** in an index plot or vs. risk score.

Checking Functional Form of Covariates

The procedure (Therneau, Grambsch & Fleming, 1990)

- 1 Fit the **null model** (no covariates) and compute martingale residuals $r_{Mi}^{(0)}$
- 2 Plot $r_{Mi}^{(0)}$ against each covariate X_j
- 3 The shape of the scatter (with a smoother) reveals the **correct functional form** of X_j

How to read the plot:

- ▶ Positive slope (roughly linear): include X_j as a *linear* term
- ▶ Negative slope: include X_j linear, but with *negative* coefficient
- ▶ Curved pattern: consider $\log X_j$, X_j^2 , or fractional polynomials
- ▶ Roughly horizontal (flat): X_j may not be needed at all

Practical notes:

- ▶ Plots are often noisy — always superimpose a smoother (LOWESS or LOESS)
- ▶ If form of some covariates is known, fit those first; plot null-model residuals against the remaining covariates
- ▶ If $\log X_j$ is suspected, plot residuals against $\log X_j$; a straight line confirms the log transformation

Example 4.5 — Multiple Myeloma: Functional Form Check

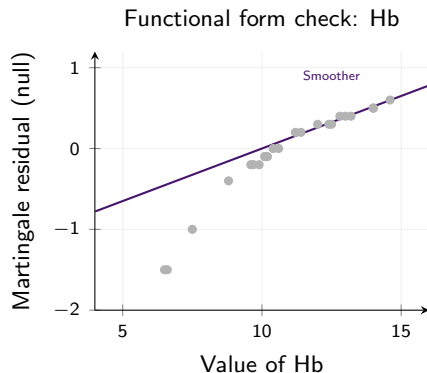
Selected model: Bun + Hb (Example 3.5)

Procedure:

- 1 Fit null model $\Rightarrow$ martingale residuals
- 2 Plot vs. Hb (Figure 4.7) and Bun (Figure 4.8)
- 3 Add LOWESS smoother

Results:

- ▶ **Hb:** plot shows a *negative linear* slope $\Rightarrow$ linear term appropriate. Higher Hb $\Rightarrow$ lower hazard.
- ▶ **Bun:** plot shows a *positive linear* slope $\Rightarrow$ linear term appropriate. Higher Bun $\Rightarrow$ higher hazard.
- ▶ **log(Bun):** smoother shows curvature $\Rightarrow$ log transformation is *not* appropriate as a linear term (Figure 4.9).



Questions?

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